DE–6174 **11**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) List out the elements of computer and communication system. Explain. (10)

   (b) Discuss the development in communication technology. (10)

2. (a) Write short notes about "Intellectual Property rights" (10)

   (b) Write the basic features of Data base software. (10)

3. (a) Explain the use of presentation graphics software. (10)

   (b) Write about the concept of Groupware. (10)

4. (a) Discuss about online information services. (10)

   (b) Explain the concept of Video Conferencing with an example. (10)

5. Write short notes on the following.

   (a) Electronic Data Interchange (10)

   (b) Cyberethics. (10)

6. (a) Explain about secondary storage devices. (10)

   (b) Write about compression and decompression. (10)

7. (a) Explain the types of Database organization. (10)

   (b) How to organise Data in secondary storage. Explain. (10)

8. (a) Describe "The Information Super Highway". (10)

   (b) Explain the six phases of system Analysis and Design. (10)

DE–6175 **12**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the binary and octal number systems in detail with examples. (10)

   (b) Describe the number codes and character codes in detail with examples. (10)

2. (a) Write in detail about basic theorems and properties of boolean algebra. (10)

   (b) What is a decoder? Also explain a 3-to-8 line decoder in detail with its truth table. (10)

3. (a) Explain the quadruple 2-to-1 line multiplex in detail with its function table. (10)

   (b) Simply the Boolean function

   $F(w,x,y,z) = \Sigma$ (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14) using four variable Karnaugh map. (10)

4. (a) Explain the full-adder in detail with its truth table. (10)

   (b) Describe the construction of arithmetic logical unit in detail. (10)

5. (a) Explain the working principle of Master-slave Flip-flop in detail with diagram. (10)

   (b) Describe the working principle of BCD counter in detail with diagram. (10)

6. (a) Write in detail about static RAM and dynamic RAM. (10)

   (b) Discuss about magnetic disk memories and magnetic tape memories. (10)

7. (a) Explain the direct memory access in detail with diagram. (10)

   (b) Discuss about the Interrupts and Interrupt priorities. (10)

8. (a) Write in detail about instruction cycle and execution cycle. (10)

   (b) Describe the sequence of control registers in detail. (10)

DE–6176 **13**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Discuss the structure of C program. (5)

   (b) What are variables? List the rules for naming a variables? (5)

   (c) Explain the branching statements in C. (10)

2. (a) What is data type? Explain the data types available in C. (10)

   (b) Discuss the following with example

      (i) Switch (4)

      (ii) Break and continue. (6)

3. (a) Define functions. How do you define a function in C? (5)

   (b) Explain any 5 library functions in C with example. (10)

   (c) How strings are stored in array? Explain with example. (5)

4. (a) What do you mean by recursion function? Explain recursion in C with example. (10)

   (b) Explain in detail about multidimensional array. (10)

5. (a) What do you mean by pointers? List the advantages of using pointers in C. (5)

   (b) Write short notes on array on pointers. (5)

   (c) Define structures. How to define a structure in C? (5)

   (d) Discuss the file modes in C. (5)

6. (a) How do you declare a pointers in 'C'? Explain with example. (5)

   (b) Explain in detail about the passing pointers to the functions. (5)

   (c) How structure members are accessed and processed? (5)

   (d) Explain with example about the functions to open and close the file. (5)

7. (a) Define stack. Discuss in detail about the stack operations and representation. (10)

   (b) Write short note on binary and decimal integers. (5)

   (c) Write short note on queue operations. (5)

8. (a) What are linked list? How nodes are inserted and removed from the linked list? (10)

   (b) Explain in detail about the threaded binary trees. (10)

DE–6177 **14**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the component of windows desktop. (10)

   (b) Discuss the managing files with windows explorer. (10)

2. (a) Explain the steps to insert date and time to a document. (10)

   (b) Explain the mail merge features of the MS-word with an example. (10)

3. (a) Discuss the formatting the text in MS-word. Explain. (10)

   (b) Write short notes on Zoom out pages in page orientation. (10)

4. (a) Explain the Excel opening screen in detail. (10)

   (b) Discuss about any four functions in MS-Excel. (10)

5. (a) Explain the method of creating charts in Excel. (10)

   (b) Explain how to sort the records in Excel database. (10)

6. (a) Explain how to use template in power point. (10)

   (b) Explain how to use Formatting Bullets and Numbering in MS Power Point. (10)

7. (a) Discuss the table manipulation in Power Point. (10)

   (b) How will you prepare Handouts? (10)

8. Write short notes on the following :

   (a) Page Numbering in MS-word. (8)

   (b) Header and Footer in Excel. (6)

   (c) Forms in Access. (6)

DE–6178 **21**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the basic concepts of object oriented programming in detail. (10)

   (b) Discuss about tokens and keywords in detail with example programs in C++. (10)

2. (a) Write in detail about benefits of object oriented programming and object oriented languages. (10)

   (b) Discuss about identifiers, variables and manipulators in C++ with examples. (10)

3. (a) Describe the 'Continue' and 'GOTO' statement in C++ in detail with example programs. (10)

   (b) Explain the multidimensional arrays in detail in C++ with example programs. (10)

4. (a) Write in detail about structures in C++ with example programs. (10)

   (b) Describe the 'DO..WHILE' and 'FOR' Loop in detail in C++ with example programs. (10)

5. (a) Discuss about argument passing using pass-by-value in function with example programs in C++. (10)

   (b) Write in detail about variables, scope and libraries with examples in C++. (10)

6. (a) Explain the concept of pointers to functions in detail in C++ with example programs. (10)

   (b) Describe the concept of virtual functions in C++ in detail with example programs. (10)

7. (a) Write in detail about hybrid inheritance and hierarchical inheritance in C++ with example programs. (10)

   (b) Describe the concept of destructors in C++ in detail with example programs. (10)

8. (a) Discuss about error handling during file operations in C++ with example programs. (10)

   (b) Write in detail about file pointers and file directions in C++ with example programs. (10)

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DE–6179 **22**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain three different types of systems. (10)

   (b) Discuss the concepts of MIS and DSS. How are they related? How do they differ? (10)

2. (a) Write a note on system development life cycle. (10)

   (b) Explain the roles played by a system analyst. (10)

3. (a) Explain the steps in feasibility analysis. (10)

   (b) Write short notes on DFD. Give an example. (10)

4. (a) Explain and illustrate the key elements of a structured chat. (10)

   (b) Write a note on decision table and decision tree. (10)

5. (a) Explain the various design methodologies. (10)

   (b) Explain the various form design. (10)

6. (a) Write a detailed account on indexed file organization. (10)

   (b) Discuss hierarchical and relational database design concepts. (10)

7. (a) Explain the role of database administration. (10)

   (b) Discuss in detail about system testing. (10)

8. (a) What is system maintenance? Explain corrective and adaptive maintenance. (10)

   (b) Discuss the major phases in hardware and software selection. (10)

DE–6180 **23**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is property window? Explain. (10)

   (b) Write short notes on event procedure. (10)

2. (a) List out the common properties of visual basic controls. (10)

   (b) Discuss the sequence of events triggered for a form. (10)

3. (a) With a suitable example program, illustrate date function. (10)

   (b) How do you pass an array to functions and procedures? Explain. (10)

4. (a) How do you use object browser to navigate user-defined functions and procedures? Discuss. (10)

   (b) Illustrate static and dynamic arrays with an example program. (10)

5. (a) Discuss briefly on procedures and sub procedures. Give example. (10)

   (b) How do you create classes in VB? Discuss. (10)

6. (a) Write short notes on built in objects in VB. (10)

   (b) Discuss briefly on MDI forms. (10)

7. (a) Write brief notes on debugging tools. (10)

   (b) With an example program, discuss animation in VB. (10)

8. (a) Discuss briefly on reading and writing statements. (10)

   (b) What do you mean by drag and drop? Give an example program. (10)

DE–6181 **24**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2013

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

($5 \times 20 = 100$)

Answer any FIVE questions,

All questions carry equal marks.

1. (a) Write in detail about computer graphics. (10)

   (b) Briefly explain circle generation algorithms. (10)

2. (a) Describe the video display generation in detail. (10)

   (b) Explain Bresenham's algorithm, (10)

3. (a) Explain composite transformations? (10)

   (b) Briefly explain

      (i) Basic Transformations and

      (ii) Matrix Representation. (5+5)

4. (a) Explain sutherland Hodgman algorithm? (10)

   (b) Explain the following (3+3+4)

      (i) Point Clipping

      (ii) Line segment clipping

      (iii) Convex polygon clipping.

5. (a) Explain Translation and Rotation. (10)

(b) Briefly explain 3D Transformations. (10)

6. (a) Write in detail about cohen sutherland algorithm. (10)

(b) Explain specifying projection plane and view volume. (10)

7. (a) Describe the Z – buffer algorithm in detail. (10)

(b) How to construct a three dimensional view? (10)

8. (a) Explain components of user unterface. (10)

(b) What are the styles of command language? (10)

DE–6182 **31**

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Construct the truth table for the following formula

$\neg(P \vee (Q \wedge R)) \rightleftarrows ((P \vee Q) \wedge (P \vee R))$ (4)

(b) Obtain the principal disjunctive normal form of

$(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$. (8)

(c) Obtain the principal conjunctive normal form of

$P \rightarrow (P \wedge (Q \rightarrow P))$. (8)

2. (a) Explain well-formed formulas. (6)

(b) Show that

$(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee \neg(x)Q(x)$. (8)

(c) Show that $S \vee R$ is tautologically implied by

$(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$. (6)

3. (a) Prove that for any two sets A and B $A-(A\cap B)=A-B$. (5)

(b) Prove that $A\cup(B\cap C)=(A\cup B)\cap(A\cup C)$ by using Venn diagram. (5)

(c) Let $R=\{\langle 1,2\rangle,\langle 3,4\rangle,\langle 2,2\rangle$ and $S=\{\langle 4,2\rangle,\langle 2,5\rangle,\langle 3,1\rangle\langle 1,3\rangle\}$ Find $R\circ S, S\circ R, R\circ R,(R\circ S)\circ R$, $R\circ(R\circ R)$. (10)

4. (a) Let $X=\{1,2,3,4,5,6,7\}$ and $R=\{\langle x-y\rangle / x-y$ is divisible by $3\}$. Show that R is an equivalence relation. Draw the graph of R. (8)

(b) Show that the functions $f(x)=x^3$ and $g(x)=x^{1/3}$ for $x\in R$ are inverses of one another. (8)

(c) Prove that $\sim\sim A=A$ by using characteristic function ψ_A. (4)

5. (a) Define semi group homomorphism and Monoid homomorphism with examples. (6)

(b) Prove that for any commutative monoid $\langle M,*\rangle$, the set of idempotent elements of M forms a submonoid. (6)

(c) Let $f:R\to R$ be defined by $f(x)=2x-3$. If f is $1-1$ and onto then find f^{-1}. (8)

6. (a) Define group and give example. (4)

(b) State and prove Lagrange's theorem. (10)

(c) Determine all the proper subgroups of the symmetric group $\langle S_3,*\rangle$. (6)

7. (a) Define the following terms :

(i) Undirected edge

(ii) Simple graph

(iii) Length of the path

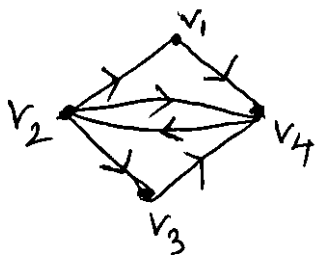
(iv) Cycle

(v) Tree. (10)

(b) Prove that in any graph, the number of vertices of odd degree is even. (8)

(c) Define a digraph. (2)

8. (a) Obtain the adjacency matrix A of digraph given below : (6)



(b) Prove that a graph with n vertices is a tree. (8)

(c) Prove that a graph G with n vertices is a tree if and only if it has $(n-1)$ edges and no cycle. (6)

DE–6183 **32**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

OPERATING SYSTEMS

(2005 – 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the history of operating systems. (10)

   (b) What are system calls? Explain. (10)

2. (a) What is a process? Explain Race conditions in process management. (10)

   (b) Explain :

      (i) Monitors.

      (ii) Event counters. (10)

3. (a) What are Device controllers? Explain. (10)

   (b) What are functions of Interrupt Handlers? (10)

4. (a) Explain the goals for I/O software. (10)

   (b) What are the ways to prevent and avoid deadlocks? Explain. (10)

5. (a) Write a detailed note on memory management with Bit-maps and linked lists. (10)

   (b) Explain segmentation with suitable diagrams. (10)

6. (a) Write a detailed note on memory management without swapping or paging. (10)

   (b) What is segmentation in virtual memory? Explain with suitable diagrams. (10)

7. (a) What is thrashing? Discuss. (5)

   (b) What is the purpose of resource allocation graph? How are the used in deadlocks? (5)

   (c) What are interrupts? Discuss the use of interrupt handlers for the various classes of interrupts. (10)

8. (a) Explain the different disk scheduling algorithms in detail. (10)

   (b) (i) What is a file?

   (ii) What are directories?

   (iii) What is a file server?

   (iv) Discuss the protection mechanisms for files. (10)

DE–6184 **33**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

RDBMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is a database management system? Explain its features. (10)

   (b) Write a detailed note on schema and subschema for databases. (10)

2. (a) What is a relational database model? Explain with an example. (10)

   (b) Write short notes on :

      (i) Data manipulation (5)

      (ii) Data control facilities (5)

3. (a) Explain the benefits of oracle with examples. (10)

   (b) Describe the oracle architecture with suitable illustrations. (10)

4. (a) Write a detailed note on client/servers systems. (5)

   (b) Explain oracle queries with examples. (10)

   (c) Write a note on subqueries. (5)

5. (a) Write a detailed note on DML commands. (10)

   (b) What are use of unions and multiple part queries? Explain with suitable examples. (10)

6. (a) Write a note on network system tables in oracle. (10)

   (b) What are indexes? Explain in detail with examples. (10)

7. Explain the following with suitable examples :

   (a) Create table statement (5)

   (b) Modifying tables (5)

   (c) Renaming a table (5)

   (d) Dropping a table (5)

8. (a) What are the languages supported by oracle precompiler? Explain. (10)

   (b) Write a detailed note on embedded SQL statements. (10)

DE–6185 **34**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005–2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain in detail about Inodes with example. (10)

   (b) Write note on directory level permissions. Explain about changing of permission. (10)

2. (a) (i) Give an account on command line structure. (5)

   (ii) List out the shell meta characters. (5)

   (b) Explain in detail about shell variables with examples. (10)

3. (a) What is the command for each of the following tasks? Explain each command with proper example.

   (i) Replacing a file. (5)

   (ii) Killing processes by name. (5)

(b) Explain the following :

(i) awk Built–in variables. (5)

(ii) awk operators. (5)

4. (a) Discuss in detail about the pick and news commands with proper example. (10)

(b) Explain in detail about while and until Loops in shell programming with example. (10)

5. (a) Explain about Debugging in unix. (10)

(b) (i) Write note on accessing the environment in UNIX. (5)

(ii) Explain about File – I/O (read and write). (5)

6. (a) Give a detailed account on standard I/O Functions with explanation. (10)

(b) List out and explain character set macro with example. (10)

7. (a) Discuss about Built–In functions. (10)

(b) Explain in detail about control flow and relational operators. (10)

8. (a) How functions and procedures are used in programming? (10)

(b) Explain the following :

(i) mm formatting commands. (5)

(ii) troff special character sequences. (5)

DE–6186 **35**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Third Semester

COMMUNICATION SKILLS

(2005–2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Explain the modes of verbal and non-verbal communication. (10)

   (b) What are the four skills essential for an effective communication? Explain. (10)

2. (a) What is Academic listening? (10)

   (b) Write a conversation between two persons about the details of a course. (10)

3. (a) What are the methods of writing meeting-minutes? (10)

   (b) Describe skimming and scanning. (10)

4. (a) How does spoken communication differ from written communication? (10)

   (b) What are the strategies for good conservation? (10)

5. (a) What are the various types of questions that are frequently asked in interviews? (10)

   (b) Write a self-Introduction that you would make to a H.R. at Interview. (10)

6. (a) How to choose your career? (10)

   (b) What are pre-interview preparation? Explain. (10)

7. (a) Differentiate a C.V/Resume and a bio-data. (10)

   (b) How to make an effective presentation? (10)

8. (a) How does presentation help to understand the audience? (10)

   (b) What are the various types of reports? (10)

DE–6187 **41**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the following :

   (i) Uses of Computer Network. (5)

   (ii) Network hardware and Network Software. (5)

   (b) OSI reference model. (10)

2. (a) Explain how twisted pairs can be used as transmission media. (12)

   (b) Compare and contrast satellite communication with fiber optic communication. (8)

3. (a) (i) What are the design issues in data link layer? (5)

   (ii) What are the error detection and correcting codes in datalink layer? (5)

   (b) Discuss the physical layer transmission media with suitable diagrams. (10)

4. (a) Discuss about Adhoc Network Routing. (12)

   (b) Discuss about ICMP. (8)

5. (a) Explain in detail about RARP. (12)

   (b) Write a detail notes on Internet multicasting. (8)

6. (a) Explain in detail about TCP. (13)

   (b) Explain in detail about UDP. (7)

7. (a) Discuss about the flow control and buffering. (10)

   (b) Explain protocols for Gigabit Network. (10)

8. (a) Explain in detail about secret and public key algorithm. (10)

   (b) Discuss about WWW and electronic mail. (10)

DE–6188 **42**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. From the following Trial Balance of Shri.Amandeep Singh prepare his final accounts for the year ended on December 31, 2003. (20)

Particulars	Dr.	Cr.
Capital	–	1,00,000
Drawings	18,000	–
Buildings	15,000	–
Sales	–	1,00,000
Bank Balance	20,000	–
Furniture & Fittings	7,500	–
Motor van	25,000	–
Loan from Hari @ 12% Interest	–	15,000
Interest paid on the above	900	–
Purchases	75,000	–
Opening Stock	25,000	–
Establishment Expenses	15,000	–

Particulars	Dr.	Cr.
Wages	2,000	–
Insurance	1,000	–
Commission Received	–	4,500
Sundry Debtors	28,100	–
Sundry Creditors	–	10,000
Interest	–	3,000
Total	2,32,500	2,32,500

Adjustments :

(a) The value of Stock on 31.12.2003 was Rs.32,000.

(b) Outstanding wages Rs.5,000

(c) Prepaid Insurance Rs.300

(d) Commission Received in advance Rs.1300

(e) Allow interest on capital @ 10%

(f) Depreciate building at 2.5%, Furniture and Fitting 10% and Motorvan 10%

(g) Charge interest on drawings Rs. 500

(h) Accrued interest Rs. 500.

2. From the following particulars pertaining to assets and liabilities of a company calculate (20)

(a) Current Ratio

(b) Quick Ratio

(c) Proprietary Ratio

(d) Debt – Equity Ratio and

(e) Capital Gearing Ratio.

Liabilities	Rs.	Assets	Rs.
5,000 Equity Shares of Rs. 100 Each	5,00,000	Land and Building	6,00,000
2,000 8% Preference share of Rs. 100 each	2,00,000	Plant and Machinery	5,00,000
4,000 9% Debentures		Stock	2,40,000
of Rs. 100 each	4,00,000	Debtors	2,00,000
Reserves	3,00,000	Cash and bank	50,000
Creditors	1,50,000	Pre-paid	
Bank overdraft	50,000	expenses	10,000
	16,00,000		16,00,000

3. Define the term 'Cost'. Explain the various classification of Cost. (20)

4. From the following particulars of M/S D.D. Dewan & Co. prepare stores ledger account showing the pricing of material issues under (20)

   (a) FIFO method and

   (b) LIFO method.

02-08-2010	Opening stock	800 units at 4.20
03-08-2010	Purchased	800 units at 4.20
04-08-2010	Issued	1,200 units
06-08-2010	Purchased	1,600 units at 4.80
07-08-2010	Issued	1,000 units
09-08-2010	Purchased	400 units at 6.00

11-08-2010	Issued	800 units
13 -08-2010	Issued	100 units
15-08-2010	Purchased	500 units at 8.00.

5. Discuss in detail about various Accounting Concepts and Conventions. (20)

6. From the following data forecast the cash position at the end of April, May and June 1998. (20)

Month	Sales Rs.	Purchases Rs.	Wages Rs.	Sundry expenses Rs.
February	1,20,000	80,000	10,000	7,000
March	1,30,000	98,000	12,000	9,000
April	70,000	1,00,000	8,000	5,000
May	1,16,000	1,03,000	10,000	10,000
June	85,000	80,000	8,000	6,000

Further information :

(a) Sales : 10% realized in the month of sales. Balance equally realized in two subsequent months.

(b) Purchases : Creditors are paid in the month following the month of supply.

(c) Wages : 20% paid in arrears in the following month.

(d) Sundry expenses : Paid in the month itself.

(e) Income tax : Rs.20,000 payable in June.

(f) Dividend : Rs.12,000 payable in June.

(g) Income from investments Rs.2,000 received half-yearly in March and September.

(h) Cash balance on hand as on 1.4.98 Rs.40,000.

7. The sales turnover and profit during two years were as follows : (20)

Year	Sales (Rs.)	Profit (Rs.)
1991	1,40,000	15,000
1992	1,60,000	20,000

Calculate :

(a) P/V Ratio.

(b) BEP in volume.

(c) Sales required to earn a profit of Rs.40,000.

(d) Fixed expenses.

(e) Margin of safety when sales are Rs. 1,20,000.

8. Compute cost per running kilometers from the following data of a Mini Bus. Estimated life of the vehicle is 1,00,000 kms. Annual running is 25,000 kms. (20)

Particulars	Rs.
Cost of vehicle	8,00,000
Road license (annual)	10,000
Insurance (annual)	5,000
Garage rent (annual)	6,000
Supervision and salaries (annual)	1,18,000

Particulars	Rs.
Cost of fuel per litre	40
Repairs and maintenance per KM	5
Tyres allocation per KM	2

Charge interest at 5% per annum on cost of vehicle and it runs 20 kms. per hour on an average and one litre of fuel gives 10 km.

DE–6189 **43**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Solve the following LPP graphically (20)

Maximize $Z = 4x_1 + 3x_2$

Subject to the constraints :

$$2x_1 + x_2 \le 1{,}000$$
$$x_1 + x_2 \le 800$$
$$x_1 \le 400 \text{ and } x_2 \le 700$$
$$x_1 \ge 0 \text{ and } x_2 \ge 0.$$

2. Solve the following Assignment problem which minimizes the total man hours. (20)

		Men			
		E	F	G	H
Task	A	18	26	17	11
	B	13	28	14	26
	C	38	19	18	15
	D	19	26	24	10

3. Define queueing system and define the elements of queueing system briefly. (20)

4. Obtain the steady state solution for the model $(M/M/1):(\infty/FIFO)$. (20)

5. What is simulation? What are the elements of simulation model? How to generate Random numbers? (20)

6. With this notation construct a network for the following :

   $A < D, A < E, B < F, D < F, C < G, C < H, F < I, G < I$.

   Also find the minimum time of completion of the project when the time (in days) of completion of each task is as follows. (20)

Task :	A	B	C	D	E	F	G	H
Time:	8	10	8	10	16	17	18	14

7. The cost of new machine is Rs. 5,000 The maintenance cost of nth year is given by $C_n = 500(n-1); n = 1,2,....$ suppose that the discount rate per year is 0.5. After how many years it will be economical to replace the machine by a new one? (20)

8. A small project is composed of seven activities whose time estimates are listed in the table as follows:

Activity		Estimated duration		
i	j	Optimistic	Most likely	Pessimistic
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

(a) Draw the project Network.

(b) Find the expected duration and variance of each activity. What is the expected project length?

(c) Calculate the variance and standard deviation of project length. What is the probability that the project will be completed?

DE–6190 **44**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the evolution of the object model in detail. (10)

   (b) Describe the elements of the object model in detail. (10)

2. (a) Write in detail about the nature of an object. (10)

   (b) Discuss about the relationships among objects. (10)

3. (a) Explain the nature of the class in detail. (10)

   (b) Describe the relationship among classes in detail. (10)

4. (a) Write in detail about the importance of proper classification. (10)

   (b) Discuss about the key abstraction mechanism. (10)

5. (a) Discuss about the UML. (10)

   (b) Explain the purpose of analysis and design in detail. (10)

6. (a) Write in detail about the concept of patterns. (10)

   (b) Describe the concept of Use-cases in detail with examples. (10)

7. (a) Describe the essentials of class diagram in detail. (10)

   (b) Write in detail about the class diagram with examples. (10)

8. (a) Describe the interaction diagram in detail with examples. (10)

   (b) Write in detail about the activity diagram with examples. (10)

DE–6191 **45**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Discuss about the concept of Java in windows. (10)

   (b) Explain the Microsoft internet implementation in detail. (10)

2. (a) Write in detail about the Javascript. (10)

   (b) Describe the concept of VB script in detail. (10)

3. (a) Discuss about the concept of CGI and perl. (10)

   (b) Discuss about an overview of Internet programming. (10)

4. (a) Explain the Netscape Extensions in detail. (10)

   (b) Describe the Shokwave and Lingo in detail. (10)

5. (a) Write in detail about VDO live Technology. (10)

   (b) Describe the concept of pulling web information in detail. (10)

6. (a) Discuss about the creating Netscape Navigator Plug–Ins. (10)

   (b) Explain the creation of an Activex Control to Activate a webpage in detail with an example. (10)

7. (a) Explain the Adding graphics to web pages in detail with examples. (10)

   (b) Describe the site and page design in detail. (10)

8. (a) Discuss about the graphic view of web. (10)

   (b) Write in detail about dynamic graphics and animation. (10)

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DE–6192 **51**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the elements of digital geometry in detail. (10)

   (b) Explain about the motivation and perspectives of image processing. (10)

2. (a) Discuss about the texture and pattern discrimination in image processing. (10)

   (b) Explain the applications of digital image processing. (10)

3. (a) Discuss about the contrast intensification and smoothing. (10)

   (b) Discuss about the low-pass and high-pass filtering. (10)

4. (a) Write in detail about Homomorphic filtering. (10)

   (b) Describe about restoration by maximum a posterior estimation. (10)

5. (a) Explain about loss-less compression. (10)

   (b) Write the detail about the run-length coding. (10)

6. (a) Discuss about details of plane-to-plane transformation mapping. (10)

   (b) Write in detail about the processing of color images. (10)

7. (a) Write in detail about any one of the segmentation method in the image processing. (10)

   (b) Describe in detail about the threshold selection methods. (10)

8. (a) Discuss about the pattern filtering approach. (10)

   (b) Describe the elements of extraction by thresholding. (10)

DE–6193 **52**

DISTANCE EDUCATION

M.C.A./ M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define software engineering. Explain about the development process model. (10)

   (b) Describe the linear sequential model and rapid application development (RAD) model in detail with diagrams. (10)

2. (a) Explain about the software matrics in detail with examples. (10)

   (b) Discuss about Role of management in software development. (10)

3. (a) Discuss about estimating software scope, Resources, and project estimation. (10)

   (b) Discuss about software project planning. (10)

4. (a) Explain about the programming Team structure. (10)

   (b) Describe about the planning for quality assurance. (10)

5. (a) Write in detail about software cost estimation model. (10)

   (b) Explain the software configuration management. (10)

6. (a) Explain the information flow in detail with diagrams. (10)

   (b) Write in detail about specification analysis. (10)

7. (a) Describe the details about structure specifications Techniques. (10)

   (b) Explain about the architectural design. (10)

8. (a) Discuss about the software testing methods. (10)

   (b) Discuss about the testing strategies in software engineering. (10)

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DE–6194 **53**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Fifth Semester

VISUAL PROGRAMMING – II

(2005-2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about Hungarian notation with example. (10)

   (b) Write a Small windows program to display "Hello" message. (10)

2. (a) Describe the mainframe object in detail. (10)

   (b) What are the files created by appwizard for windows programs? Explain. (10)

3. (a) How do you add Caret to the windows? Explain. (10)

   (b) Explain about Mouse event messages. (10)

4. (a) What are the three types of menus in a windows application? Explain. (10)

   (b) How do you add, remove and modify menu items during execution? (10)

5. (a) Explain about windows MessageBox( ) function. (10)

(b) What are the two types of Dialog boxes? Explain. (10)

6. (a) Write a VC++ program for creating simple calculator application. (10)

(b) Explain the steps to create a simple Notepad in VC++. (10)

7. (a) Discuss in detail about MDI programs. (10)

(b) Explain the sequential and random file implementation in VC++. (10)

8. (a) What is an Exception? What are the exception classes in VC++? (10)

(b) Explain the OLE feature in VC++. (10)

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DE–6195 **54**

DISTANCE EDUCATION

M.C.A./M.C.A.(Lateral) DEGREE EXAMINATION, MAY 2013.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

($5 \times 20 = 100$)

Answer any FIVE questions.

All questions carry equal marks.

1. (a) Write detailed notes on compiler writing tools. (10)

   (b) Explain about design of lexical analyzer. (10)

2. (a) Discuss shift reduce parsing and its implementation. (10)

   (b) Explain the steps involved in operator precedence parsing. (10)

3. Write an overview about

   (a) Three-address code (5)

   (b) Quadruples (5)

   (c) Triples (5)

   (d) Postfix notation (5)

4. Give detailed description about syntax directed translation. (20)

5. (a) Write notes on contents of symbol table. (10)

   (b) Discuss about procedure for representing scope information. (10)

6. Describe about principal source of optimization. (20)

7. Write an overview about the DAG representation of basic blocks with an illustration. (20)

8. How to create a simple code generator? Discuss. (20)

DE–6196 **55**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the concepts of data mining. (10)

   (b) Explain how data mining is used in marketing. (10)

2. (a) Discuss about the data selection, cleaning and enrichment process of knowledge discovery in databases. (10)

   (b) Write a detailed note on genetic algorithms. (10)

3. (a) Write short notes on Association rules. (10)

   (b) Discuss about Neural Networks. (10)

4. (a) Describe the architecture of data warehouse. (10)

   (b) Write the functions of warehouse manager. (10)

5. (a) Define (i) aggregations (ii) data marting. (10)

   (b) Write notes on : (i) database scheme, (ii) meta data. (10)

6. (a) Explain the server hardware design of a data warehouse. (10)

   (b) Discuss about 'operating the data warehouse'. (10)

7. (a) Explain about the client hardware design of a data warehouse. (10)

   (b) Discuss about the capacity planning of a data warehouse. (10)

8. (a) What do you mean by tuning the data warehouse? (10)

   (b) Explain how to test a data warehouse for its correct functionality. (10)

DE–6197 **61**

DISTANCE EDUCATION

M.C.A/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2013.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain in about the stepwise planning activities. (10)

   (b) Explain about the project analysis and technical planning. (10)

2. (a) Explain cost benefit analysis and cost benefit evaluation techniques. (10)

   (b) Explain in detail cocomo model. (10)

3. (a) Discuss in detail about the projects and activities. (10)

   (b) How can you identify the critical activities? Explain. (10)

4. (a) Write the strategies for risk reduction. (5)

   (b) How to visualize the progress of the project and explain various methods available for visualizing the project progress. (15)

5. (a) What do you mean by network planning models? How it is formulated? (10)

(b) What are the objectives of activity planning? (5)

(c) How critical path is identifying in network model? (5)

6. (a) Explain the various schedules in resource allocation. (15)

(b) Explain change control. (5)

7. (a) Discuss in detail about the basic functions and responsibilities. (10)

(b) Explain about prototyping model in detail. (10)

8. (a) What are the components of PRINCE project? (5)

(b) Explain PRINCE project organization and procedures. (10)

(c) Write short note on PRINCE project stage boundaries. (5)

DE–6198 **62**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the benefits and problems in multimedia in detail. (10)

   (b) Write in detail about system components. (10)

2. (a) Briefly explain multimedia commercial tools and standards. (10)

   (b) Describe multimedia platforms. (10)

3. (a) Explain compression techniques in images. (10)

   (b) Explain analog, digital audio/video in detail. (10)

4. (a) Describe the JPEG and DVI in detail. (10)

   (b) Explain video compression techniques. (10)

5. (a) Explain multimedia objects and classes. (10)

   (b) Briefly explain multimedia frameworks. (10)

6. (a) Explain CD-family and CD-i. (10)

   (b) Describe multimedia on networks. (10)

7. (a) Discuss about redundancy and visibility. (10)

   (b) Explain evaluating a compression system. (10)

8. (a) Explain media types. (10)

   (b) Describe the training and education using multimedia in detail. (10)

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DE–6199 **63**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Briefly explain artificial neural networks? (10)

   (b) Briefly explain activation functions in neural networks. (10)

2. (a) Describe the biological neural networks in detail with diagram. (10)

   (b) Explain neuron models. (10)

3. (a) Explain the Boltzmann machine concept in detail. (10)

   (b) Discuss about supervised and unsupervised learning. (10)

4. (a) Explain single layer perceptrons. (10)

   (b) Discuss counter propagation networks. (10)

5. (a) Explain BAM. (10)

(b) Explain Hopfield networks. (10)

6. (a) Explain the overview of adaptive resonance theory. (10)

(b) Discuss optical neural network. (10)

7. (a) Explain Hebbian learning. (10)

(b) Briefly explain learning tasks. (10)

8. (a) Explain about perceptrons. (10)

(b) Describe the back propagation training algorithms. (10)

DE–6200 **64**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2013.

Sixth Semester

C SHARP (C #)

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

All questions carry equal marks.

1. (a) Write a note on common language run time. (10)

   (b) Discuss the components of .Net framework. (10)

2. (a) Describe the concept of error handling in detail. (10)

   (b) Write a note on VOS type system. (10)

3. (a) Discuss in detail about intermediate language and meta data. (10)

   (b) Write about common language specification in C#. (10)

4. (a) With an example, explain the concept of Enumeration. (10)

   (b) What is events? Demonstrate with an example program. (10)

5. (a) Define constructor. Discuss types of constructor with example coding. (10)

   (b) Write a C# program to demonstrate the concept of boxing and unboxing. (10)

6. (a) Explain the various iteration statements with an example. (10)

   (b) Write in detail about DO's and Don'ts of exception handling. (10)

7. (a) How do you create a simple client application? Discuss with an example. (10)

   (b) Explain the concept of Debugging and setting Breakpoints. (10)

8. (a) Discuss in detail about intermediate language Disassembler. (10)

   (b) Explain about platform invocation services. (10)

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DE–6632 **15**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

First Semester

LAB I — DATA STRUCTURES USING "C"

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiners should select and give ONE question to each candidate by lot system.

1. (a) Write a C program to sort a set of elements using selection sort.

   (b) Write a menu program in C to find an element using linear and binary search method.

---------- Cut here ----------

2. (a) Write a C program to sort a set of elements using insertion sort.

   (b) Write a C program for linked list implementation of queue operations.

3. (a) Write a C program to find the roots of a quadratic equation

$aX^2 + bX + c = 0.$

(b) Write a C program for matrix manipulation (addition, subtraction, multiplication using function).

---------- Cut here ----------

4. (a) Write a C program to convert a given number into words for number 1 to 5, 1 as ONE, 2 as TWO and 5 as FIVE.

(b) Write a C program to find the roots of a quadratic equation using function definition

$aX^2 + bX + c = 0.$

---------- Cut here ----------

5. (a) Write a C program to implement all string operations using switch statement.

(b) Write a C program to find the factorial of a given number using recursion.

---------- Cut here ----------

6. (a) Write a C program to copy contents of one file to another file.

(b) Write a C program to evaluate the given mathematical expression using stack.

7. (a) Write a C program to implement insert and delete operations on linked list structure.

   (b) Write a C program to sort 10 nos. in ascending order and display the value before and after sorting.

---------- Cut here ----------

8. (a) Write a C program to implement all string operations using switch statement.

   (b) Write a C program to read 10 values to an array variable and use pointers to locate and display each value.

DE–6633 **16**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Break-up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Answer any TWO questions choosing ONE from Section – A and another from Section – B.

Each questions carries 50 marks.

SECTION – A

1. Type a lecture note of your own and provide audio explanation with the help of sound files.

---------- Cut here ----------

2. Prepare your resume with your photograph inserted. Use table, bullets and different color features.

3. Create a new document and type the following mathematical expression :

$$\int_{\infty}^{\infty} xdy \left| dx \sum_{i=1}^{100} ax_i + b + |A| \pm \sqrt[3]{ax + b^2} \right. .$$

---------- Cut here ----------

4. Create a work sheet in Ms-Excel with following columns :

Employee Number, Name, Designation, Basic Pay, HRA, DA, LIC, PF, Gross pay, Net pay.

(a) Type data for emp no, emp Name, designation, Basic pay, LIC and PF.

(b) Calculate :

HRA = 20% of Basic pay

DA = 25% of Basic pay

Gross pay = Basic pay + HRA + DA

Net pay = Gross pay - (LIC + PF).

(c) Draw the bar chart between emp name and net pay.

(d) Sort the designation column and employee name column at a time.

5. The following were the observations made in certain experiments for the values y and given the value of x.

x :	1	2	3	4	5	6	7	8	9	10
y :	10	30	45	25	15	28	40	32	15	35

Compute the following :

(a) Find the Mean, Median and standard deviation.

(b) Correlation coefficient between x and y.

(c) Draw the Bar chart and shading cells.

(d) Draw a chart in Excel and paste it an word.

---------- Cut here ----------

6. Crate two tables SALES 1, SALES 2 and join them to produce a third table SALES 3.

---------- Cut here ----------

7. Create inventory table with fields item No, Name, Quantity on hand, reorder level.

(a) Create a form in custom format

(b) Create a query REORDER to show all the items

(c) Create a Macro that executes reorder query automatically.

8. Create a student table with the following fields Reg No., name, Mark 1, Mark 2, Mark 3. Create address table with fields Reg No., Street, City, and Pin.

   (a) Write a query to display Reg No., Name and total of all the three subject Marks.

   (b) Write a query to display Reg No., Name, Street, City, Pin and total of all the three subject Marks.

---------- Cut here ----------

SECTION – B

9. Create HTML page to display 10 courses and their highlighted features, conducted by an educational institution in a colorful format.

---------- Cut here ----------

10. Create HTML page that plays video and audio files.

---------- Cut here ----------

11. Create HTML page to display the pictures of elephant and Monkey. Links to another page when user clicks overt the picture. The linked page should display features of the animal clicked.

DE–6634 **25**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

Second Semester

LAB III — GRAPHICS USING C++

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer TWO questions ONE from Section A and another from Section B chosen by the candidates through Lot system.

Write C++ programs and execute them.

Break-up marks

Record Note : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

SECTION A

1. Write a C++ program to show graphical transformation (Translation, Rotation) of a two dimensional object.

---------- Cut here ----------

2. Write a C++ program to draw a circle using simple DDA technique.

3. Write procedure and a C++ program to draw non-linear object circle and three Concentric Circle.

---------- Cut here ----------

4. Write a C++ program to manipulate the graphical object using SHADING technique.

---------- Cut here ----------

5. Write a C++ program to draw non-linear objects arcs and curve.

---------- Cut here ----------

SECTION B

6. Draw any 3-D object and perform animation on it.

---------- Cut here ----------

7. Write a program to draw the object bi-cycle and rotate the wheels through animation.

---------- Cut here ----------

8. Develop a program to do the following :

   (a) Draw a triangle ABC, translate it to a distance and then rotate it through an angle clockwise.

   (b) With the same triangle, reverse the operations as in case (a).

9. Implement the polygon clipping algorithm using C++ graphics mode graphics function.

---------- Cut here ----------

10. Draw a circle. Divide into four quadric and fill up different color for each quadrant.

DE–6635 **26**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2013.

Second Semester

LAB IV — VISUAL BASIC

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each sub division carries 50 marks.

Mark break up :

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Write a VB project that receives a year number from a textbox and month Name from listbox and displays number of days in the given month. Take care of leap years use lost focus event for list box.

   (b) Write a VB project that stores 10 students record with fields stu no, stu name, age, sex and consolidated percentage of marks in an array display data.

2. (a) Write a VB project that stores 10 employee records with fiedlds empno, Nam, Age, Sex and salary in an array. Display data fields in textboxs and provide command buttons to move to desired record.

(b) Write a VB project to screen saver that displays a list of pictures with 1 second pause in between successive pictures.

---------- Cut here ----------

3. (a) Design and implement one ActiveX control for stop watch used for sports.

(b) Design and implement and ActiveX control for railway clock (hours display 1 to 24).

---------- Cut here ----------

4. (a) Write a VB project to display the available of seats when the user selects the train name and date from the Combobox and timings from the another Combobox.

(b) Write a VB project to display the availability of seats when the user selects the bus name and number from the Combobox and date and time from another Combobox.

---------- Cut here ----------

5. (a) Create a VB application to make the textbold, italic, underline and also to color the text (Using check box and option button).

(b) Create an application using VB to move elements from one list to another list and add new items.

6. (a) Write a VB project to display a message (Example left/right mouse button is clicked) when the mouse button is clicked using mouse move event of the form.

(b) Design and implement one ActiveX control for monthly calendar.

---------- Cut here ----------

7. (a) Write a VB project using AxDLL or EXE add a class module that would perform the following function

(i) Sum of digits

(ii) Whether the given number is Amstrong or not.

(b) Write a VB project using control array that creates a scientific calculator with appropriate command buttons, include the following capabilities for the calculator : +, –, *, /, % power, square root, square and log (base 10).

---------- Cut here ----------

8. (a) Write a VB project for a blood bank that maintains a list of donors with address and their blood group provide the following reporting features

Display all donors (using data reports)

(i) In age group 20 – 30

(ii) In a particular city.

(b) Write a VB project to calculate the amount when the user select the class type of the train from the Combobox and also enters the number of passengers.

9. (a) Write a VB project to display the availability of seats when the user selects the flights name from the combo box and data including timings from another combo box.

(b) Design and implement on ActiveX control for digital electric current usage measurement meter used in houses.

---------- Cut here ----------

10. (a) Write a VB project for blood bank that maintains a list of donors with address and their blood group provide the following reporting features

Display all the donors (using data reports)

(i) With particular blood group

(ii) Male donors with particular blood group

(iii) Female donors with particular blood group.

(b) Write a VB project for commercial bank operations using SB account database with the following features :

(i) ADD NEW ACCOUNT

(ii) Calculate simple interest and update balance taking average of last 6 month balance in the account

(iii) Close account.

DE–6636 **36**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Third Semester

LAB V — RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Students has to answer ONE question by lot system.

1. A hospital maintains a list of blood donors with the following fields :

   Donor number, Donor name age, Donor address, Blood group (A, B, AB & O).

   Write a program using oracle to print out the donor number, name and address of the donors.

   (a) having A blood group

   (b) donors with age above 25 years.

———————————————— cut here ————————————————

2. A Daily sales file contains records with the following fields :

   Item number, date, item description, price for each item, quantity, cost of each item.

   Write a program using oracle to calculate the total amount of sales and profit for each sale.

3. Write a program in SQL to prepare a salary report for 10 employees.
   Assume the following format for files.
   Eno, Name, Basic Pay (BP), Allowances, Deductions, Gross pay (GP), Net pay.
   (DA) Dearness Allowance = 65% of BP
   Gross pay = BP + DA
   Deduction = P.F. + Loan
   Calculate IT based on the following condition :

Salary	IT
< = 2 lakh	Nil
> 2 lakhs, < = 3 lakhs	10%
> 3 lakhs	20%

   Total deduction = Ded + IT.

---------- cut here ----------

4. A news paper vendor buys newspaper from a distribution for Rs. 4 and sells them in retail for Rs. 5. Prepare a report in the following format for 10 weeks.

Week	Bought	Sold	Profit	Loss
1.				
2.				
Total				

---------- cut here ----------

5. An examination was conducted for 10 students with the following inputs.

   Reg. No., name, 4 test marks. The minimum pass for each subject is 50. The following grading system.

Average	Grade
0–49	Fail
50–65	C
66–74	B
75–100	A

   Write an SQL program to do the following. Assign a letter grade to each student based on the average score and list regno, average score and grade.

6. Write a program in SQL for Hospital Billing System with the following fields :

   Patient no, Patient name, Doctor, type (IN/OUT), charges for consultation, blood test charge, x-ray charges and total fee.

   Prepare reports with the following conditions :

   (a) The patients undergone blood test

   (b) The patients who taken X-rays.

———————————————— cut here ————————————————

7. Write an SQL program for Canteen Maintenance with two tables namely Menu and BILL.

   The MENU table contains the fields ITEM, RATE.
   The following items are available : Tea and Coffee, Idly
   BILL Table contains empno, name, date of issue, no. of item 1, rate 1, no. of item 2, rate 2 and total amount.
   Rate = Unit rate * no. of items
   Total amount = Rate 1 + Rate 2.

   Prepare a report as given in the format given below.

S. No.	Emp. No.	Name	Purchased Rate
1			
2			
3			
		Total amount	

———————————————— cut here ————————————————

8. Write a PL SQL program to prepare a mark statement using student table and mark table. Assume your own data.

———————————————— cut here ————————————————

9. Write a PL SQL program for library information system to print the number of books, title of the book, accession number. Assume your own data.

10. Prepare a banking statement for printing the transaction for a given customer. Print the date, type of transaction (credit, debit) and total balance at the end of a month. Assume your own data.

DE–6637 **37**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Third Semester

LAB VI — ALGORITHMS AND SHELL PROGRAMMING

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Breakup of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Students has to answer ONE question by lot system.

1. (a) Write shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located. (Using UNTILL statement).

   (b) Write a program in C/C++ for Linear and Binary search and execute the program.

2. (a) Write a program in C/C++ for Quick sort using divide and conquer algorithm.

(b) Write a shell program using 5 arguments to take the pattern as well as input and output file names. If the pattern is found, display the message "Pattern Found" else display "Pattern Not Found". Also check the right number of arguments and print the error message if number of arguments is not equal to five.

---------- Cut here ----------

3. (a) Write a program in C/C++ for Heap sort.

(b) Write a shell program to prepare electric bill for domestic customers.

For First 100 units — Rs. 1.00 / unit

For Next 100 units — Rs. 2.50 / unit

Above 200 units- Rs. 4.00 /unit

Prepare the electrically bill in the following format.

Customer Number: _____

Customer Name : _____

Previous Reading : _____

Current Reading : _____

Units Consumed : _____

Charge : _____

---------- Cut here ----------

4. (a) Write a shell program to sum up the series.

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+.....+\frac{1}{20!}.$$

(b) Write a mean driven shell program to copy, edit, rename and delete a file.

5. (a) Write a program in C/C++ for straight exchange sorting.

(b) Write a shell program which accepts the name of a file from the standard input and perform the following tests on it

(i) File readable

(ii) File writable

(iii) File existence

(iv) Both readable and writable.

---------- Cut here ----------

6. (a) Write a Menu driven shell program for the following.

(i) List of Files.

(ii) Processes of Users.

(iii) Today's Date

(iv) Users of System.

(v) Quit to UNIX.

(b) Write a program in C/C++ to find factorial for a given number.

---------- Cut here ----------

7. (a) Write a program in C/C++ for Quick sort using divide and conquer algorithm.

(b) Write a menu driven shell program to copy, edit, rename and delete a file.

8. (a) Write a shell program for fore function calculator by using yacc command.

(b) Write a program in C/C++ to solve knight tour problem using backtracking method.

---------- Cut here ----------

9. (a) Write a program in C/C++ to perform Ackerman's Function.

(b) Write a menu driven shell program to perform the following task

(i) Enter the sentence in a file

(ii) Search a given whole word in the existing file

(iii) Quit.

---------- Cut here ----------

10. (a) Write a program in C/C++ for travelling salesman problem using branch and bound method.

(b) Write and execute a shell program to generate leave status report of an organization.

DE–6638 **46**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Break up of marks 50
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Students has to answer one question by lot system.

Each subdivision carries 50 marks.

1. (a) Write a Java program using Applet to display 3 images when 3 buttons in the border layout are clicked. The image should be displayed at the center.

   (b) Write a swing program to create the tabbed panels.

---------- Cut here ----------

2. (a) Using alteast 20 HTML tags, create a screen with the string "Web Design".

   (b) Write a menu driven Java program to do database functions using database connectivity facility. Include display, edit, insert and delete a record, functions from a table.

3. (a) Write a Java program using Applet to display the different colors and fonts.

   (b) Write a VB script to do the following :

      (i) Display the current date.

      (ii) Find the difference between the two dates.

      (iii) Find the age of a person by providing date of birth.

      (Use the date in dd/mm/yy format).

---------- Cut here ----------

4. (a) Design and publish a web page for a college.

   (b) Write a VB script to create a calender for given month and year.

---------- Cut here ----------

5. (a) Write a Java program using Applet to display the dialog menu in applet.

   (b) Write a Java program to create a color as the background.

---------- Cut here ----------

6. (a) Develop a picture gallery having at least 5 pages. Each of them is having several pictures with suitable information.

   (b) Write a Java script to create a window by using the confirm message.

---------- Cut here ----------

7. (a) Write a Java script to create a order form to select the house articles.

   (b) Write a Java program using Applet to create the frames and its controls.

8. (a) Write a VB script to do the following.

   (i) Check the given password

   (ii) Change the existing password.

   (b) Write a Java script to create a color palet and display the text in the color chosen from the palet with background color.

---------- Cut here ----------

9. (a) Write a swing program to create buttons with

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Short cut key.

   (b) Develop a web page for recruitment agency in an IT industry.

---------- Cut here ----------

10. (a) Develop a single page advertisement for a shop to be opened shortly.

    (b) Create a web page in the format of front page of a news paper using text links. Alight the text with colors.

DE–6639 **47**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Fourth Semester

LAB VIII — NETWORK LAB

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Students has to answer ONE question by lot system.

Break-up of marks :

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Write a Java program to add two numbers using RMI client server technique.

   (b) Write a Java program to setup echo server and echo client.

---------- Cut here ----------

2. (a) Write a Java program to find factorial of given number using RMI client server technique.

   (b) Write a Java program to setup chat server and chat client.

3. (a) Write a Java program to find factorial of given number using FACTORIAL INTERFACE technique.

   (b) Write a Java program to display the users and their IP addresses logged in the network.

---------- Cut here ----------

4. (a) Write a Java program to ENCRYPT given word.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

5. (a) Write a Java program to setup Time Server and Time Client.

   (b) Write a Java program to read a file from the remote system.

DE–6640 **56**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Fifth Semester

LAB IX — VISUAL C++

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-up of marks :

Record Note Book	5
Algorithm/Flow chart	15
Program	35
Debugging and Execution	35
Result	10
Total	100

1. (a) Write visual C++ win 32 application program using MFC that creates a listbox and displays name of the subjects in the current semester.

   (b) Write visual C++ win 32 application program using MFC that displays current mouse coordinate in status bar.

2. (a) Write visual C++ win 32 application program using MFC that allows the user to draw pictures with the help of mouse as a free hand drawing tool.

   (b) Write visual C++ win 32 application program using MFC that displays the message "GOOD MORNING" wherever the user clicks the mouse on the client area.

---------- Cut here ----------

3. (a) Write visual C++ win 32 application program using MFC that fills the background of the client area with a bit map.

   (b) Write visual C++ win 32 application program using MFC that displays menu. Choose the menu items using keyboard accelerator keys and display appropriate messages for the selected command in message box.

DE–6641 **57**

DISTANCE EDUCATION

M.C.A (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Fifth Semester

LAB X — COMPILER DESIGN

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Examiners should select and give ONE question to each candidate by lot system.

Break-up of marks :

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. Write a 'C' program to accept a string and check whether it satisfies the following production rule (by using shift reduce parsing)

   $E \to E + E \mid E - E \mid E * E \mid E/E \mid id \mid const \mid (E)$.

---------- Cut here ----------

2. Write a 'C' program to construct the left-most derivation and right-most derivation for the following grammar.

   $S \to a \mid \wedge \mid (T)$

   $T \to T,S \mid S$

   (a) $(a,(a,a))$

   (b) To find the LEADING and TRAILING sets.

3. Write a 'C' program to construct predictive parsing table for the following production rule.

$S \rightarrow iCtSS1 \mid a$

$S1 \rightarrow eS \mid C$

$C \rightarrow b.$

---------- Cut here ----------

4. Write a 'C' program to accept a string and check whether it satisfies following production rule (By using operator precedence grammar)

$S \rightarrow AbB$

$A \rightarrow aBb$

$B \rightarrow a \mid Bba.$

---------- Cut here ----------

5. Write a 'C' program to construct of an SLR parsing table for the following grammar.

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id.$

---------- Cut here ----------

6. Write a 'C' program to remove spaces, comment lines, new lines and produce line numbers.

---------- Cut here ----------

7. Write a 'C' program to convert the infix expression $(a+b)*(b-c)/(a+b)$ into postfix notation, three-address code and quadruples.

8. Write a 'C' program for the following :

   (a) Identifies the keywords

   (b) Identifies the identifiers.

---------- Cut here ----------

9. Write a 'C' program to convert the infix expression $(a+b)*(a-b)/(a+b)$ into quadruples, triples and indirect triples.

---------- Cut here ----------

10. Write a 'C' program to accept a string and check whether it belongs ambiguous grammar or not by using the following production rule

$E \rightarrow E+E \mid E-E \mid E*E \mid E/E \mid id \mid const \mid (E)$.

DE–6642 **65**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2013.

Sixth Semester

LAB XI — MULTIMEDIA APPLICATIONS

(2005 to 2010 calendar year onwards)

Time : Three hours Maximum : 100 marks

Break up marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner has to choose the ONE question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program for setting up a pattern for multi-grey-level images or bilevel display using limbs algorithm with the following 2×2 matrix

$$\begin{matrix} 0 & 2 \\ 3 & 1 \end{matrix}$$

Also print the dot patterns produced on a 16 grey-level system for a pixel.

(b) Do the image animation using Flash.

2. (a) Write a C++ program to create a second image array model from the given image using grey-level transformation bunching technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

Also draw histograms for the First and Second images.

(b) Write a C++ program for 2 dimensional image animation computer graphics.

---------- Cut here ----------

3. (a) Write a C++ program to find the edges of the following image using sobel edge detection procedure.

0	0	0	0	0	2	2	0	3	3
0	0	1	0	0	0	2	4	2	2
0	2	0	0	2	4	3	2	3	3
0	1	3	3	4	4	3	3	3	3
0	1	0	4	3	3	2	4	3	3
1	0	0	1	2	3	4	2	3	4

(b) Design a sales advertisement for a product using Flash.

4. (a) Write a C++ program to compress the image compression using contour coding technique. Given image

1	1	1	1	1	1	1	1	2	2	2	2	2	2	2
1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	2	2	2	2

(b) Prepare a multimedia presentation showing the features of the software packages developed by you.

---------- Cut here ----------

5. (a) Write a C++ program for grey level transformation techniques

(i) Thresholding

(ii) Bunching

(b) Design an animated cartoon picture.

6. (a) Design a birth day greeting page including various graphic features in Flash.

(b) Write a C++ program for creating an image from the given image using automatic selection of grey level for splitting technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

---------- Cut here ----------

7. (a) Write a C++ program to find the best grey level for splitting the following image.

4	5	1	7	4	2	6	3	4	2	7	4	6	7
3	7	8	2	7	6	5	4	3	3	7	2	4	3
4	5	9	9	9	7	8	7	6	5	5	4	1	2
0	0	0	3	4	2	2	3	1	1	4	6	6	5
7	3	2	4	5	1	6	2	3	4	5	6	7	9

Using automatic selection for splitting technique.

(b) Prepare a multimedia presentation shows the features of a software packages developed by you.

8. (a) Write a C++ program for contribution using given template.

Image					Template	
1	1	3	3	4	1	0
1	1	4	4	3	0	1
2	1	3	3	3		
1	1	1	4	4		

(b) Prepare a multimedia presentation shows the features of a software packages developed by you.

---------- Cut here ----------

9. (a) Design a sales advertisement for a product using Flash.

(b) Write a C++ program for setting up a pattern for multi-grey-level images or bilevel display using limbs algorithm with the following 2×2 matrix.

$$\begin{matrix} 0 & 2 \\ 3 & 1 \end{matrix}$$

Also print the dot patterns produced on a 16 grey-level system for a pixel.

10. (a) Write a C++ program for grey level transformation techniques

(i) Thresholding

(ii) Bunching.

(b) Do the image animation using Flash.